

PROFESSIONAL INSTALLATION GUIDE

Server Operating Systems and Virtual Machine Configuration

Coverage

Ubuntu Server installation and verification, Windows Server Evaluation preparation and installation summary, BIOS vs. UEFI comparison, Ubuntu boot process, and a comparative review of Windows Server, Ubuntu Server, and Rocky Linux.

Prepared for: Server Administration / Virtualization Activity

Platform: Oracle VirtualBox

Documentation date: August 2026

Note: Screenshots in this guide are based on the installation and configuration steps completed during the activity. Where a Windows Server installation screenshot was not yet captured, the guide clearly identifies the step as a procedure or summary rather than claiming completion.

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The guide is organized to document the actual virtual-machine work completed, the required research components, and the evidence needed for submission.

1. Project Overview

This project documents the deployment and basic administration of server operating systems in Oracle VirtualBox. The main hands-on work focused on installing Ubuntu Server, configuring its network and storage, creating an administrative user, enabling OpenSSH, and verifying the running SSH service. A separate Windows Server Evaluation virtual machine was also prepared for installation, including a 40 GB virtual disk and the required installation workflow.

The activity also covers foundational server concepts: BIOS versus UEFI firmware, the Ubuntu/Linux boot process, and the differences among Windows Server, Ubuntu Server, and Rocky Linux. The resulting guide combines configuration evidence, verification commands, comparison tables, troubleshooting notes, and screenshots into one professional record.

Documentation Evidence

The following screenshots are incorporated throughout the guide as evidence of the virtual-machine setup and Ubuntu installation:

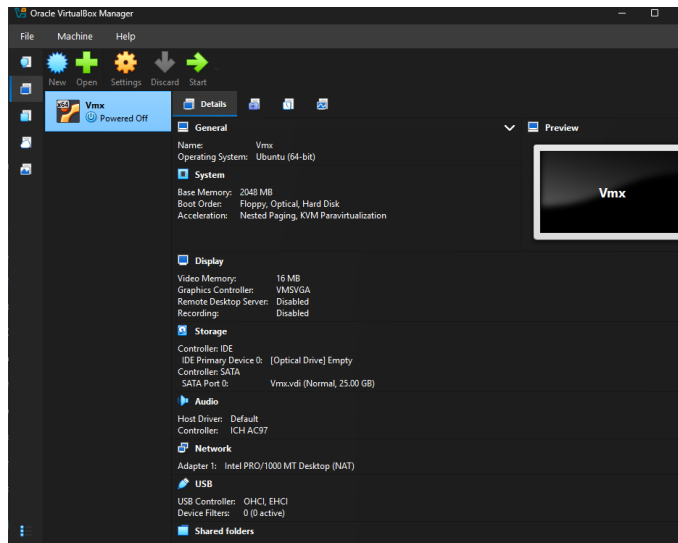


Figure 1. Ubuntu VM overview in Oracle VirtualBox.

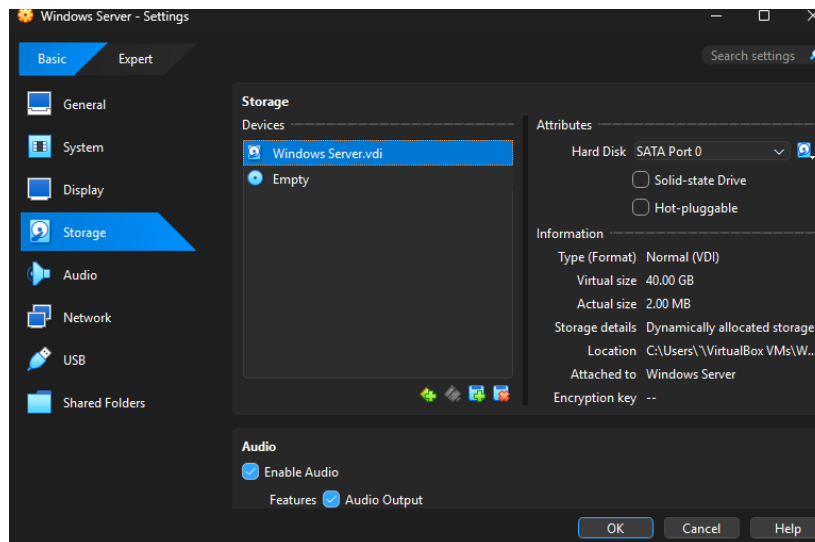


Figure 2. Windows Server VM storage configuration showing a 40 GB virtual disk.

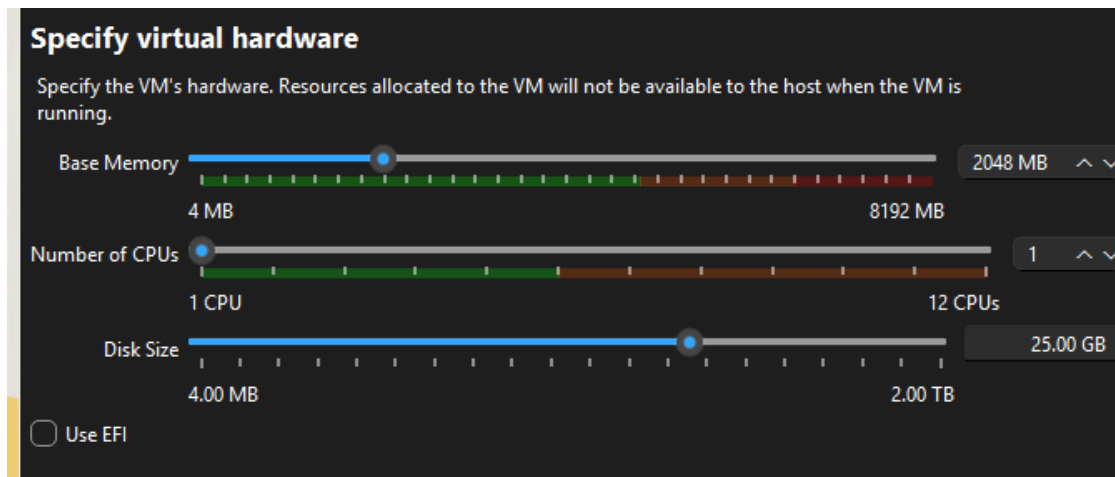


Figure 3. Virtual hardware allocation screen.

2. Objectives

The activity aims to develop practical skills in server operating-system installation, virtual-machine configuration, and basic system administration.

- Install Ubuntu Server in a new virtual machine using guided storage and DHCP network configuration.
- Assign the required hostname and create a non-root administrative user.
- Document disk partitioning, filesystem usage, disk size, and network addressing.
- Enable and verify the OpenSSH server and confirm that it is actively listening on port 22.
- Prepare and document a Windows Server Evaluation virtual machine and its installation workflow.
- Explain the differences between BIOS and UEFI and describe the Ubuntu boot sequence from power-on to login.
- Compare Windows Server, Ubuntu Server, and Rocky Linux in terms of licensing, administration, security, performance, and enterprise use.
- Record problems encountered during installation and the solutions used to resolve them.

Expected Outcome

At completion, the student should be able to install and verify a basic server operating system, perform fundamental configuration tasks, interpret service status output, and explain why different server platforms are selected for different enterprise requirements.

3. Software Requirements

Software / Resource	Purpose
Oracle VirtualBox	Creates and manages the virtual machines used for the server installations.
Ubuntu Server 26.04 LTS ISO	Installation media for the Ubuntu Server virtual machine.
Windows Server Evaluation ISO	Installation media for the Windows Server Evaluation virtual machine.
Windows host computer	Physical host used to run VirtualBox and the virtual machines.
Internet / NAT networking	Provides DHCP addressing and Internet access for package updates and server connectivity.

ISO and VM Preparation

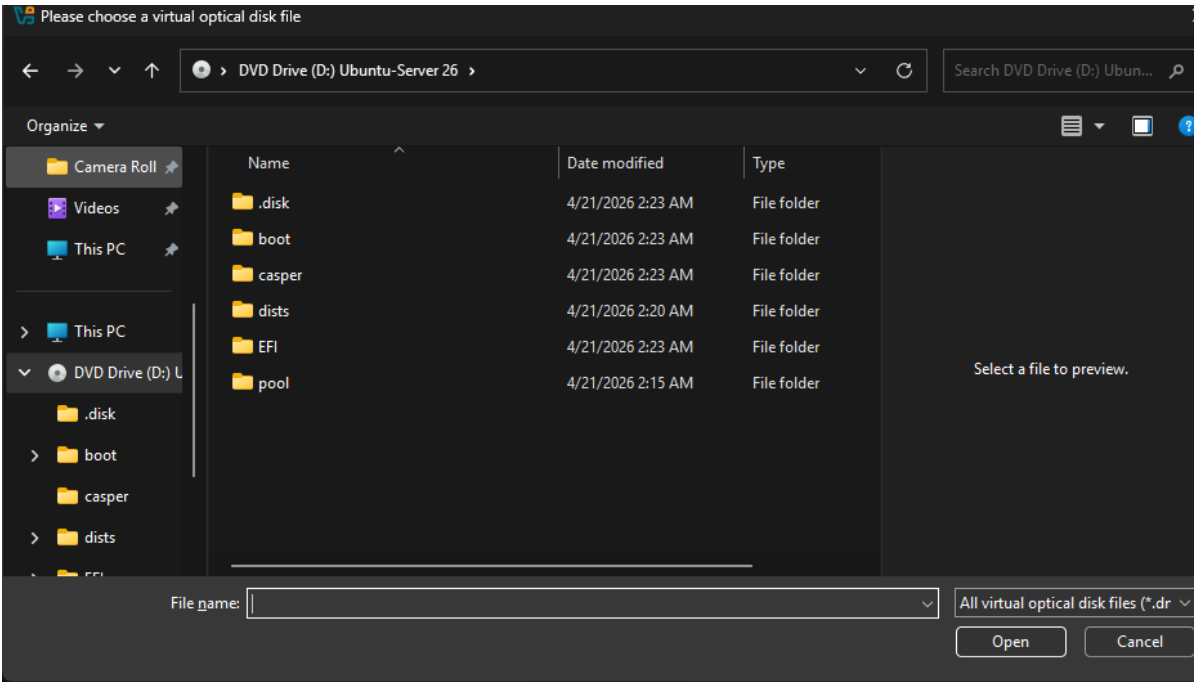


Figure 4. Virtual optical disk selection window used to attach the Ubuntu Server ISO.

The Ubuntu ISO was attached to the VM optical drive after the initial boot failure indicated that no bootable optical medium was selected. After attaching the ISO, the VM was able to start the Ubuntu Server installer.

4. Virtual Machine Specifications

Specification	Ubuntu Server VM	Windows Server VM
Virtual disk	25 GB VDI used during the recorded installation.	40 GB VDI configured.
Memory	4 GB recommended for the activity; the recorded hardware screen shows 2048 MB before adjustment.	4 GB recommended for the activity.
CPU	2 virtual CPUs recommended for the activity.	2 virtual CPUs recommended.
Network	Intel PRO/1000 adapter using NAT; DHCP assigned 10.0.2.15/24 in the recorded session.	NAT recommended for installation and updates.
Optical drive	Ubuntu Server ISO attached during installation.	Windows Server Evaluation ISO to be attached before installation.

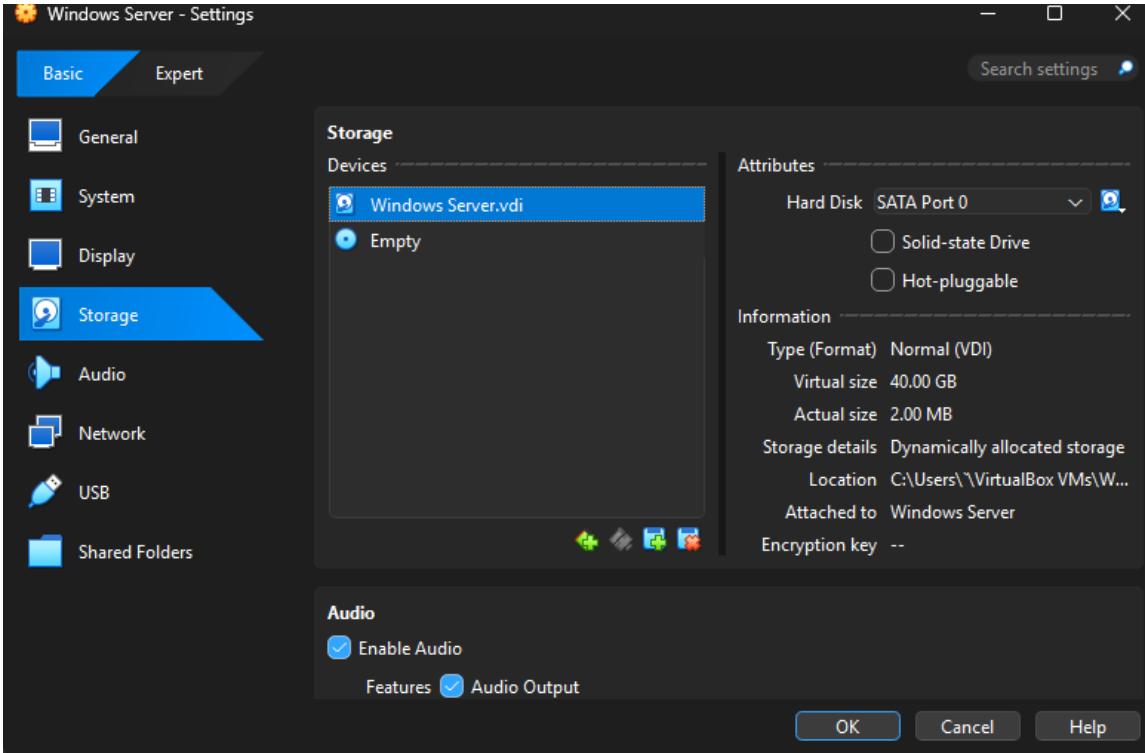


Figure 5. Windows Server VM storage configuration with a 40 GB virtual disk.

5. Step-by-Step Installation Procedure

5.1 Create and Configure the Ubuntu VM

Create a new VirtualBox VM, select Ubuntu Server 64-bit, allocate appropriate memory and CPUs, and create the virtual disk. Attach the Ubuntu Server ISO to the optical drive before starting the VM.

5.2 Start the Installer

Power on the VM and proceed through the Ubuntu Server installer. Select English and the appropriate keyboard layout. Use DHCP for the network unless the activity specifies a static configuration.

5.3 Record Network Information

The recorded installation displayed interface `enp0s3` with DHCPv4 address `10.0.2.15/24`. Record the address shown by your own VM because DHCP addresses can differ between installations.

5.4 Configure the Hostname

Assign the hostname `server01` as required by the activity.

5.5 Create the Administrative User

Create a non-root administrative account using your chosen full name, username, and strong password.

5.6 Configure Guided Storage

Select Guided storage / Use an entire disk. The recorded installation used a 25 GB disk, LVM, an ext4 root filesystem, and a separate ext4 `/boot` partition.

5.7 Install OpenSSH

Enable Install OpenSSH Server during the software-selection stage. Do not add packages that were not required by the activity.

5.8 Complete Installation and Reboot

Allow Ubuntu to finish installing. Reboot the VM and ensure the installation ISO is detached when necessary so the VM boots from the virtual disk.

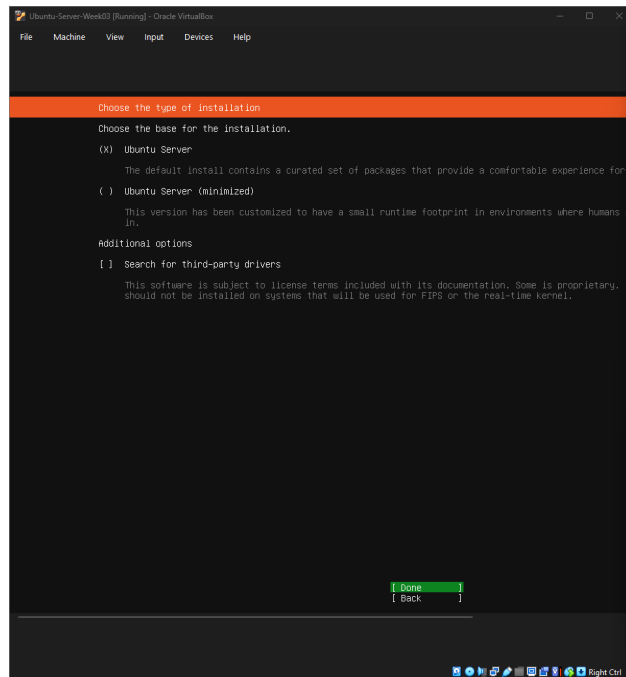


Figure 6. Ubuntu Server installation progressing through partitioning, extraction, and kernel installation.

6. Configuration Details

Network Configuration

The installer detected interface **enp0s3** and received the DHCP address **10.0.2.15/24** in the recorded session. The VirtualBox network adapter was configured for NAT. The actual DHCP address should always be recorded from the VM used for submission.

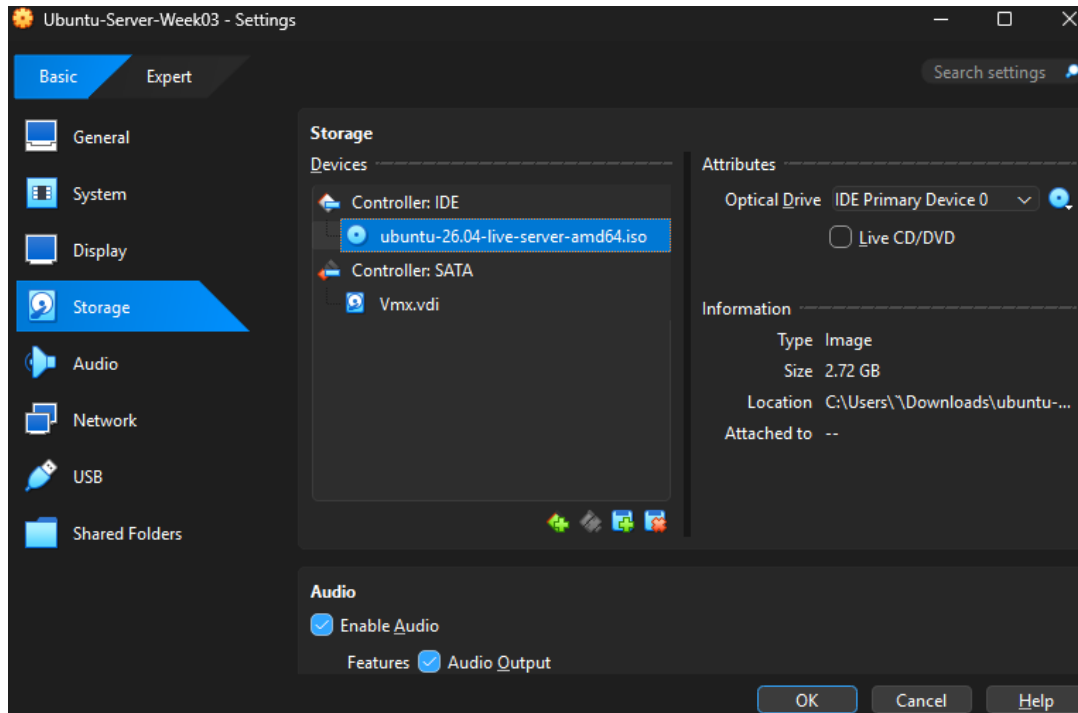


Figure 7. Ubuntu Server VM storage configuration showing the attached ISO and 25 GB virtual disk.

Storage Configuration

The guided storage configuration created a BIOS-grub partition of approximately 1 MB, a 2 GB ext4 /boot partition, and an approximately 23 GB LVM physical volume. The LVM volume group contained an approximately 11.5 GB ext4 root logical volume with approximately 11.5 GB remaining free space. This layout was accepted for the recorded activity.

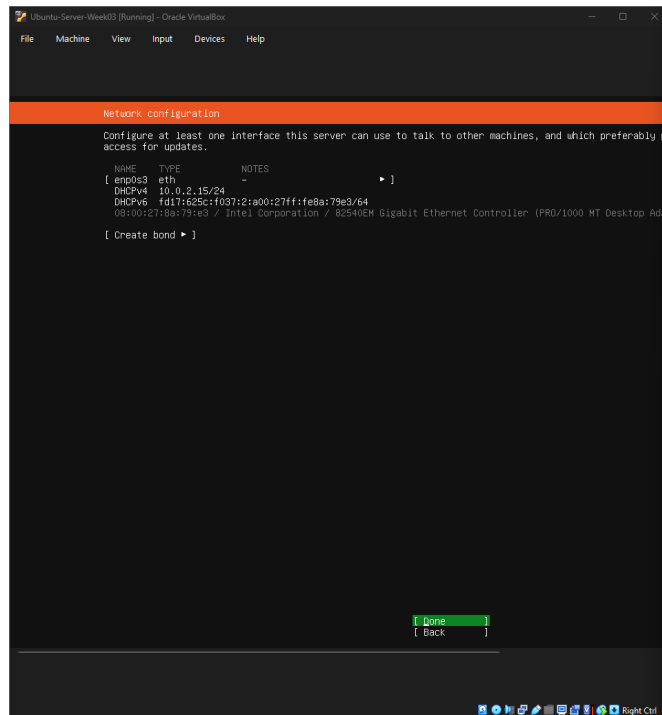


Figure 8. Ubuntu storage configuration summary showing the root and /boot filesystems and LVM layout.

7. Verification Commands

After installation, the following commands can be used to verify the basic server configuration:

Command	Purpose
systemctl status ssh	Checks the OpenSSH service status. The required result is Active: active (running) .
sudo systemctl start ssh	Starts the SSH service if it is inactive.
sudo systemctl enable ssh	Configures SSH to start automatically during boot.
hostname	Displays the current computer hostname.
ip addr	Displays network interfaces and assigned IP addresses.
lsblk	Displays disks, partitions, and logical volumes.
df -h	Displays filesystem usage in human-readable form.

```
Command: ssh - from deb ssh-client (2.9.11-2) build(2)
Try: sudo apt install <deb name>
benedict@virtualpo:~$ sudo systemctl start ssh
[sudo: authenticate] Password:
benedict@virtualpo:~$ systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-08-14 18:03:40 UTC; 15s ago
   Invocation: 3eb6b9504c6746a0a5e1725740599433
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 1422 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1425 (sshd)
     Tasks: 1 (limit: 1688)
    Memory: 1.8M (peak: 2.5M)
       CPU: 22ms
   CGroup: /system.slice/ssh.service
           └─1425 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 18:03:40 virtualpo systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 18:03:40 virtualpo sshd[1425]: Server listening on 0.0.0.0 port 22.
Aug 14 18:03:40 virtualpo sshd[1425]: Server listening on :: port 22.
Aug 14 18:03:40 virtualpo systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
benedict@virtualpo:~$ _
```

Figure 9. SSH verification showing Active: active (running) and the server listening on port 22.

8. BIOS vs. UEFI Comparison

Category	BIOS	UEFI
Definition	Traditional firmware interface used to initialize hardware and begin the boot process.	Modern firmware interface designed to replace legacy BIOS and provide a more extensible pre-boot environment.
Boot Process	Traditionally loads boot code from an MBR boot sector.	Loads UEFI boot applications from an EFI System Partition and uses a firmware boot manager.
Maximum Disk Support	Traditional MBR model has a practical boot-disk limit of about 2.2 TB.	GPT supports much larger disks; its 64-bit addressing provides a vastly larger theoretical capacity.
Partition Style	Primarily associated with MBR.	Primarily associated with GPT and an EFI System Partition.
Security Features	Limited compared with modern UEFI; no native Secure Boot in legacy BIOS.	Supports Secure Boot and signed boot components.
Speed	Generally slower and based on older initialization methods.	Designed for faster startup and modern hardware initialization.
Advantages	Mature, simple, and useful for legacy hardware and software.	Large-drive support, GPT, Secure Boot, flexible boot management, and modern firmware capabilities.
Disadvantages	Legacy capacity, partitioning, and security limitations.	Requires compatible firmware/OS configurations and can complicate legacy migrations.
Modern Usage	Mostly legacy compatibility.	Standard firmware approach on modern PCs.

Why UEFI Replaced BIOS

UEFI has largely replaced traditional BIOS because modern computers require support for larger storage devices, more flexible partitioning, stronger boot security, and newer operating systems. Traditional BIOS is commonly paired with MBR, whose addressing model creates a practical boot-disk limit of about 2.2 TB. UEFI works naturally with GPT and provides a much larger addressing space. UEFI also enables Secure Boot, which checks trusted signatures on boot software before allowing it to run. In addition, UEFI provides a more extensible pre-boot environment and integrates better with current operating-system security requirements. BIOS remains useful for legacy compatibility, but UEFI is the standard firmware approach on modern PCs.

9. Ubuntu Boot Process Flowchart

The Ubuntu boot process can be represented as a sequence from firmware initialization to a usable login prompt. The major stages required by the activity are shown below.

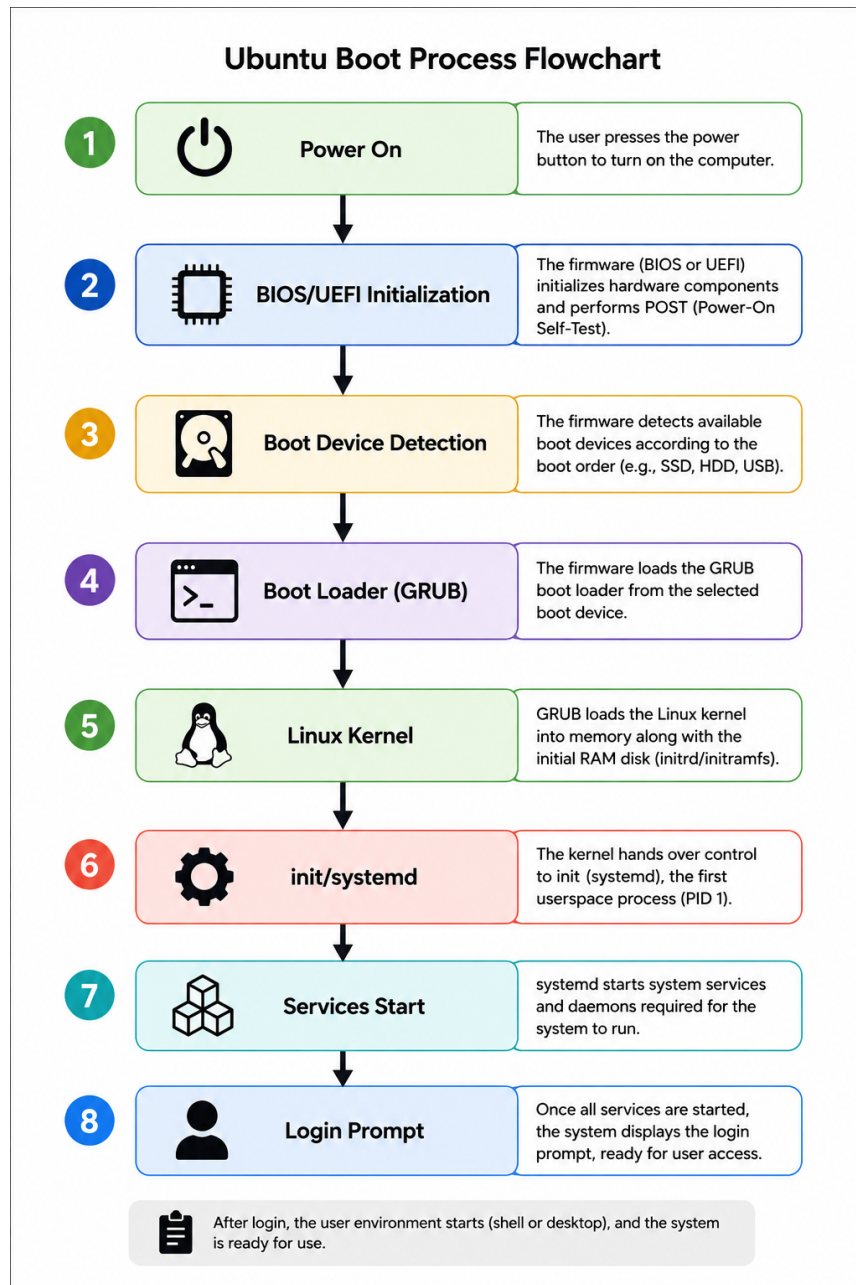


Figure 10. Ubuntu boot process: Power On → BIOS/UEFI → Boot Device → GRUB → Linux Kernel → systemd → Services → Login Prompt.

Process Explanation

- **Power On:** The computer begins the startup sequence.
- **BIOS/UEFI Initialization:** Firmware initializes hardware and performs early startup checks.
- **Boot Device Detection:** Firmware identifies a bootable device according to the configured boot order.
- **GRUB:** The GRand Unified Bootloader presents or selects the Ubuntu boot entry and loads the kernel.
- **Linux Kernel:** The kernel initializes the operating system and hardware required for userspace startup.
- **init/systemd:** systemd becomes PID 1 and manages the transition into userspace.

- **Services Start:** systemd starts required services and daemons, including services such as SSH.
- **Login Prompt:** The server becomes available for user authentication.

10. Windows Server Installation Summary

A separate Windows Server Evaluation virtual machine was prepared for the Bring-Home Assignment. The required tasks are to install the operating system, assign a computer name, create an administrator password, log in successfully, and capture screenshots of the installation.

Recorded VM Preparation

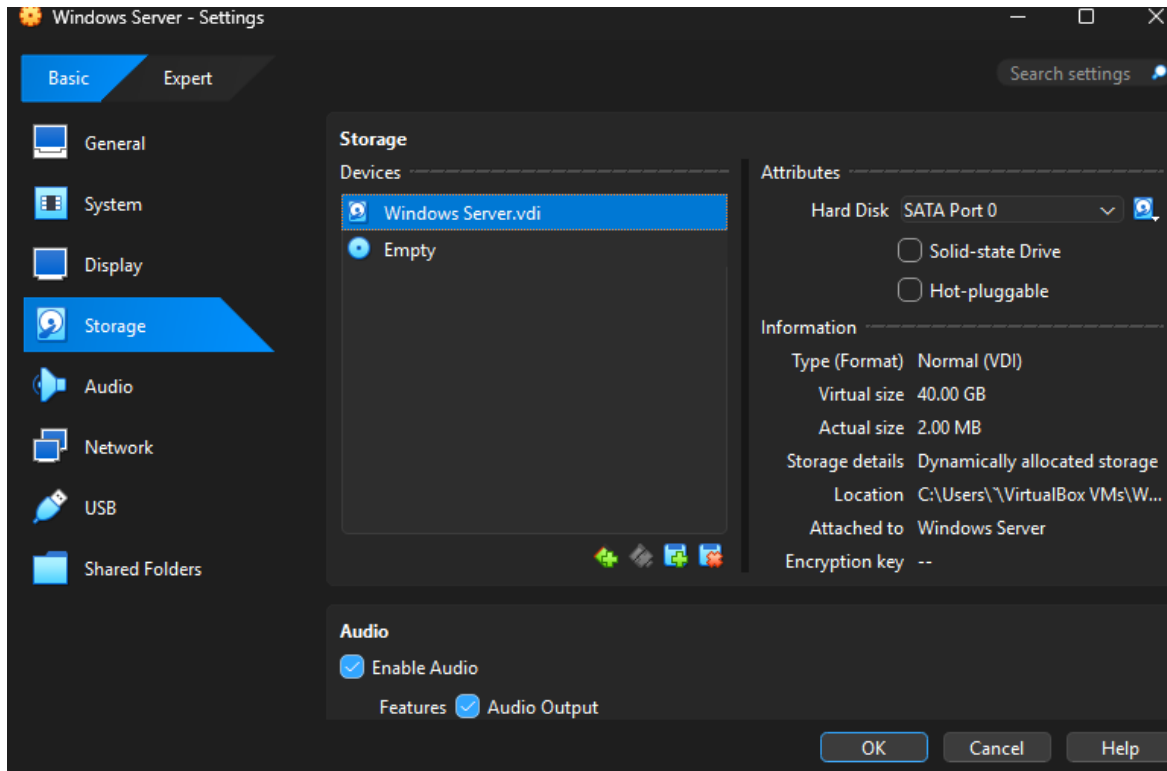


Figure 11. Windows Server VM prepared with a 40 GB dynamically allocated VDI.

Recommended Installation Procedure

- Attach the Windows Server Evaluation ISO to the virtual optical drive.
- Start the VM and select English, the appropriate time/currency format, and keyboard layout.
- Select the required Windows Server Evaluation edition. Desktop Experience is appropriate when a graphical environment is desired for the assignment.
- Accept the license terms and choose a custom installation.
- Select the unallocated 40 GB virtual disk and allow Windows Setup to create the required partitions.
- Allow Windows Setup to copy files, install features and updates, and restart as required.
- Create a strong Administrator password when prompted. Do not expose the password in screenshots.
- Log in successfully with the Administrator account.
- Open Server Manager and assign a computer name such as WIN-SERVER01, then restart if prompted.
- Capture screenshots of the installation, password setup, successful login, and computer-name configuration.

Status note: The available evidence in this guide confirms the Windows Server VM was prepared with a 40 GB virtual disk. The actual Windows installation/login screenshots should be added to the final submission after those steps are completed.

11. Operating System Comparison Table

Criteria	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Proprietary Microsoft software; Evaluation editions available for testing.	Open-source Linux; optional commercial support and security services.	Open-source Enterprise Linux distribution available at no cost.
User Interface	GUI tools such as Server Manager and Windows Admin Center; PowerShell and Server Core also available.	Primarily command-line based; GUI can be added when needed.	Primarily command-line based; Cockpit and other administration tools can be used.
Package Management	Windows application and update technologies; modern versions can use WinGet for app management.	APT and dpkg.	DNF and RPM.
Security	Active Directory, Group Policy, BitLocker, certificate services, Windows authentication, and other enterprise controls.	AppArmor, Secure Boot support, security updates, and optional extended security services.	SELinux and Enterprise Linux security practices.
Performance	Strong scalability and virtualization; generally more resource-intensive than minimal Linux installs.	Efficient and scalable, especially without a desktop environment.	Stable, predictable enterprise performance with a long support model.
Typical Enterprise Use Cases	Active Directory, Windows applications, file services, DNS/DHCP, virtualization, and Microsoft workloads.	Web servers, cloud, containers, databases, Kubernetes/OpenStack, and scale-out infrastructure.	Enterprise application/web servers, databases, infrastructure services, virtualization, and RHEL-compatible environments.
Advantages	Strong Microsoft integration, familiar GUI, centralized management, and broad commercial support.	Free/open source, large ecosystem, efficient, strong cloud support, and easy package management.	Free/open source, stable, enterprise-focused, RHEL-compatible, and suitable for production.
Disadvantages	Licensing cost, higher resource requirements, and strong Microsoft ecosystem dependence.	Some extended enterprise services require paid support; administration can require command-line skills.	Smaller ecosystem than Ubuntu and less beginner-friendly for users unfamiliar with Enterprise Linux.

Discussion

Windows Server is generally strongest where an organization depends on Microsoft technologies such as Active Directory, Group Policy, Windows applications, and centralized Windows administration. Ubuntu Server is attractive for cloud, web, container, database, and scale-out environments because of its open-source model, large package ecosystem, and efficient command-line administration. Rocky Linux is a strong choice when an organization wants an Enterprise Linux environment with RHEL compatibility and a stable, production-oriented platform.

The choice should therefore be based on workload, existing infrastructure, administrative expertise, security requirements, and budget. No single server operating system is best for every organization. Windows Server prioritizes Microsoft ecosystem integration and centralized management, Ubuntu emphasizes flexibility and broad open-source adoption, and Rocky Linux emphasizes Enterprise Linux compatibility and stability.

12. Problems Encountered and Solutions Implemented

Problem Encountered	Solution Implemented	Result
Virtual machine failed to boot because no bootable ISO was attached.	Opened VirtualBox Storage settings and selected the Ubuntu Server ISO for the optical drive.	VM successfully booted into the Ubuntu installer.
Ubuntu VM disk was 25 GB rather than the activity's stated 40 GB target.	The installation was continued using the existing 25 GB virtual disk, and the actual size was documented rather than falsely reporting 40 GB.	Installation completed; disk-size deviation recorded.
SSH service showed inactive (dead) after installation.	Ran <code>`sudo systemctl start ssh`</code> , then verified with <code>`systemctl status ssh`</code> .	SSH displayed <code>`Active: active (running)`</code> and listened on port 22.
Password input appeared invisible at the Linux login prompt.	Recognized that Linux does not display password characters while typing; entered the password normally and pressed Enter.	Login behavior understood and used correctly.
An accidental <code>`run`</code> command produced command-not-found suggestions.	Ignored the harmless command error and continued with the correct <code>systemctl</code> commands.	No system damage; SSH verification completed successfully.

13. Lessons Learned

Virtual Machine Preparation Matters

A VM must have a bootable installation medium attached before it can boot into an installer. Storage, memory, CPU, and network settings should be checked before starting.

Document Actual Configuration

If a configuration differs from the stated target, record the actual value. Accurate documentation is better than reporting a value that was not actually used.

Linux Passwords Are Invisible

Linux terminals normally do not echo password characters. This is expected behavior and is not an indication that the keyboard is malfunctioning.

Service Verification Is Essential

Installing OpenSSH is not enough by itself. `systemctl status ssh` confirms whether the service is actually active and reveals useful details such as the listening port.

LVM Provides Flexibility

Guided Ubuntu storage can use LVM, allowing logical volumes and free space to be managed more flexibly than a simple fixed-partition layout.

Firmware Affects Booting

BIOS/UEFI settings and the boot device order determine which medium the computer attempts to boot. Modern systems generally favor UEFI and GPT.

Server OS Choice Depends on Requirements

Windows Server, Ubuntu Server, and Rocky Linux each have strengths. The best platform depends on applications, management tools, security needs, compatibility, and organizational standards.

14. References

The following official documentation was used as supporting reference material for the research and technical explanations in this guide.

- **Microsoft Learn — Windows Server documentation**
<https://learn.microsoft.com/en-us/windows-server/>
- **Microsoft Learn — UEFI firmware requirements**
<https://learn.microsoft.com/en-us/windows-hardware/design/device-experiences/oem-uefi>
- **Microsoft Learn — Secure the Windows boot process**
<https://learn.microsoft.com/en-us/windows/security/operating-system-security/system-security/secure-the-windows-10-secure-boot-process>
- **Ubuntu Server documentation**
<https://documentation.ubuntu.com/server/>
- **Ubuntu Server — Package management**
<https://documentation.ubuntu.com/server/how-to/software/package-management/>
- **Ubuntu Server — OpenSSH**
<https://documentation.ubuntu.com/server/how-to/security/openssh-server/>
- **Rocky Linux documentation**
<https://docs.rockylinux.org/>
- **Rocky Linux — Licensing**
<https://rockylinux.org/legal/licensing/>
- **UEFI Forum**
<https://uefi.org/>
- **Oracle VirtualBox documentation**
<https://www.virtualbox.org/manual/UserManual.html>

Appendix — Screenshot Evidence Index

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